
Chapter 1 ~ Introduction to Digital Image Processing

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Digital Image Processing

❖ Digital

❖ Image

❖ Processing

➤ Processing of an image which is digital in nature, by a digital computer.

Motivation.....

➤ Improvement of pictorial information.

➤ Efficient storage and transmission.

What is an Image?

- ❖ Projection of 3D scene in a 2D plane is nothing but an image.

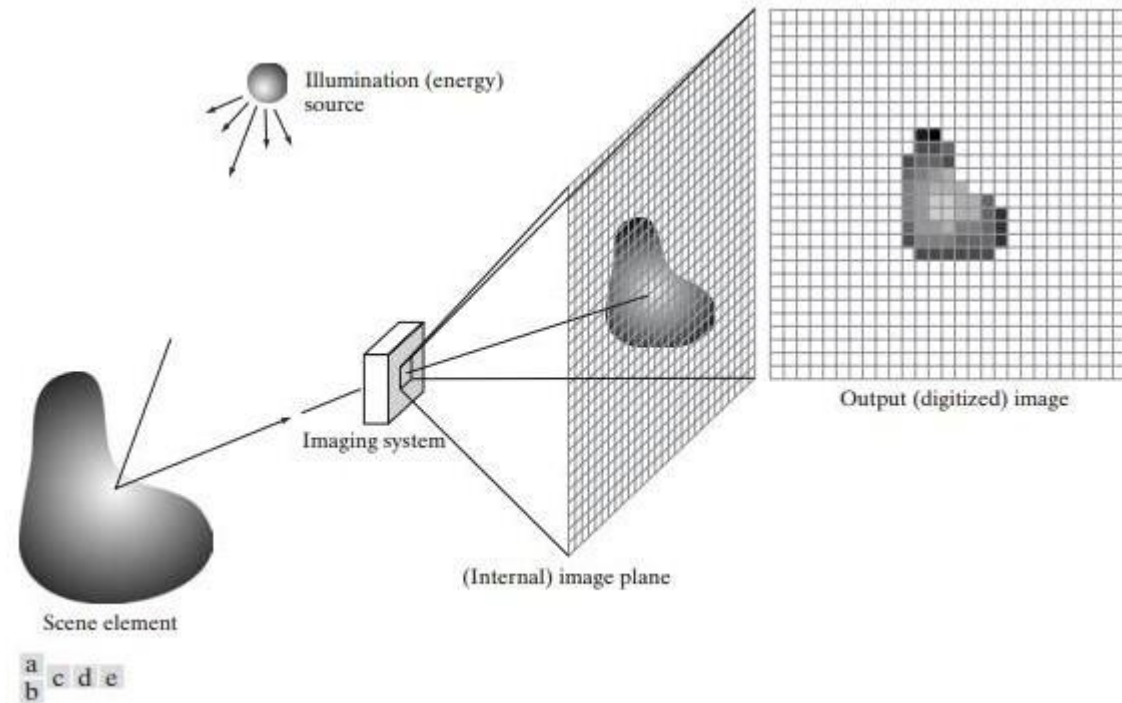


Figure: An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

What is an Image?

- ✓ **Image** is a representation of something or someone. Examples are drawing, painting and photograph etc.
- ✓ It may be **black and white** or **color**.



Figure 1: Black and white



Figure 2: Color image

Digital Image

✓ **Digital Image** represents as-

- ❖ A two dimensional function $f(x,y)$.
- ❖ Where, x and y are spatial coordinates.
- ❖ The **amplitude** of f at any pair of coordinates (x,y) is called **intensity or gray level**.
- ❖ When x, y and $f(x,y)$ all are finite and discrete quantities, we can call the image as a digital image.
- ❖ Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are referred to as-
 - i. Picture Elements
 - ii. image Elements
 - iii. Pels
 - iv. **Pixels**

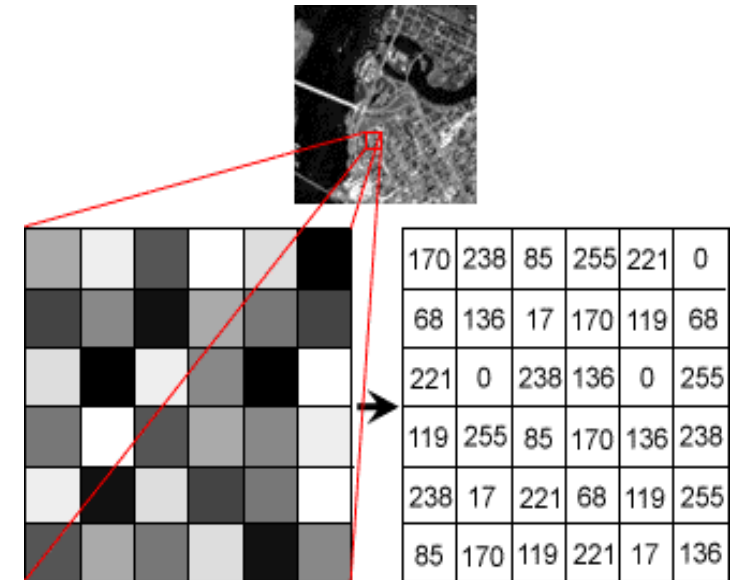


Figure 3: Digital image

Digital Image(Cont'd)

- ✓ **Digital Image** can also be represented as M rows and N columns.
- ✓ Each pixel is represented by **K bits**.
- ✓ A pixel can thus have **2^K different values**.

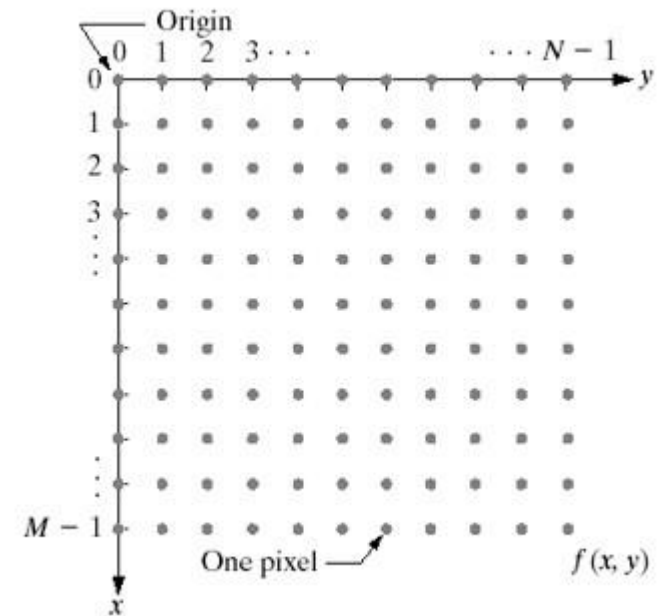


Figure 4: Row and column representation of digital image

Pixel

- ✓ The full form of the pixel is "Picture Element." It is also known as "PEL." Pixel is the smallest element of an image on a computer display, whether they are LCD or CRT monitors.
- ✓ Below is the formula to calculate the total number of pixel in an image.

$$\text{Total number of pixels} = \text{Number of rows} \times \text{Number of columns}$$

Gray Level

- ✓ The value of the minimum gray level is 0. The gray level depends on the depth of the image.
- ✓ For example: In an 8-bit image, gray level is 255. For a binary image, a pixel can only take value 0 or 255. In color image, it can choose values between 0 and 255.

Bits Per Pixel

- ❖ BPP or Bits per Pixel is used to denote the number of bits per pixel and the number depends on the depth of color.
- ❖ Image size: The size of an image depends on three things they are as following :
 - i. Number of rows
 - ii. Number of columns
 - iii. Number of bits per pixel

$$\text{Size of an image} = \text{rows} * \text{cols} * \text{bpp}$$

Image Processing

- ✓ Image processing is a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or to extract some useful information from it.
- ✓ It is a type of signal dispensation in which input is an image, and output may be image or characteristics associated with that image.
- ✓ Image processing basically includes the following three steps-
 1. Importing the image via some image acquisition tool.
 2. Analyzing and manipulating the image.
 3. Output is the last stage in which result can be altered image or report that is based on image analysis.

Purpose of Image Processing

The purpose of image processing can be divided into 5 groups-

- ✓ **Visualization**- Observe the objects that are not visible.
- ✓ **Image sharpening and restoration**- To create better image.
- ✓ **Image retrieval**- Seek for the image of interest.
- ✓ **Measurement of pattern**- Measures various objects in an image.
- ✓ **Image recognition**- Distinguish the objects in an image.

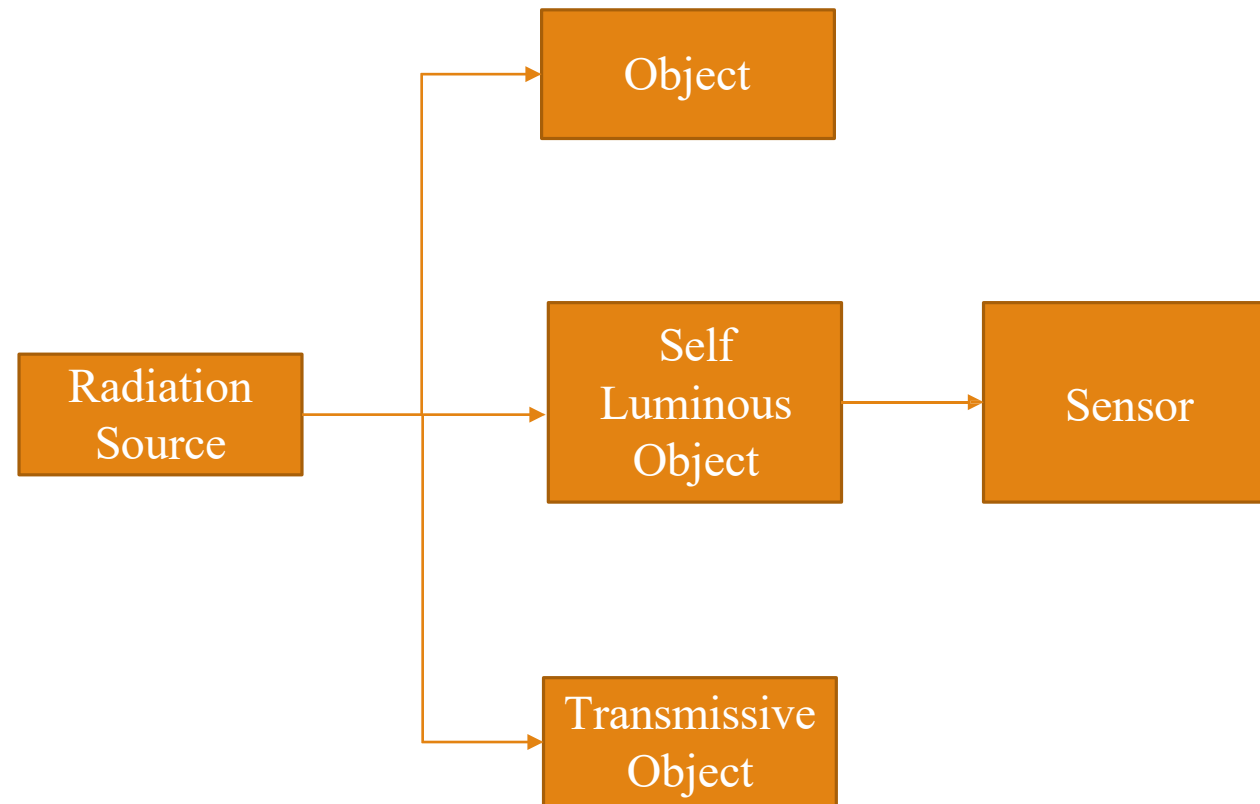
Level of Processing in DIP

- ❖ The continuum from **image processing** to **computer vision** can be broken up into low-, mid- and high-level processes.

Low Level Process	Mid Level Process	High Level Process
Input: Image Output: Image Examples: Noise removal, image sharpening	Input: Image Output: Attributes Examples: Object recognition, segmentation	Input: Attributes Output: Understanding Examples: Scene understanding, autonomous navigation

In this course we will stop here

Image Processing Environment



Types of Imaging

There are three types of based on different objects:

- i. Reflective Mode Imaging
- ii. Emissive Type Imaging
- iii. Transmissive Type Imaging

Reflective Mode Imaging

- ✓ It is the simplest form of imaging.
- ✓ It uses a sensor to acquire the digital image.
- ✓ All video cameras, digital cameras, and scanners use some type of sensors to capture the image.

Emissive Type Imaging

- ✓ Images are acquired from self luminous objects without the help of a radiation source.
- ✓ In this type of imaging the objects are self luminous.
- ✓ The radiation emitted by the object is directly captured by the sensor to form an image.
- ✓ Thermal imaging (Magnetic Resonance Imaging) is an example of emissive type imaging.

Transmissive Type Imaging

- ✓ The radiation source illuminates the object.
- ✓ The absorption of radiation by the objects depends upon the nature of the material.
- ✓ Some of the radiation passes through the objects.
- ✓ The attenuated radiation sensed to an image.
- ✓ Examples are X-Ray imaging, ultrasound imaging, microscopic imaging etc.

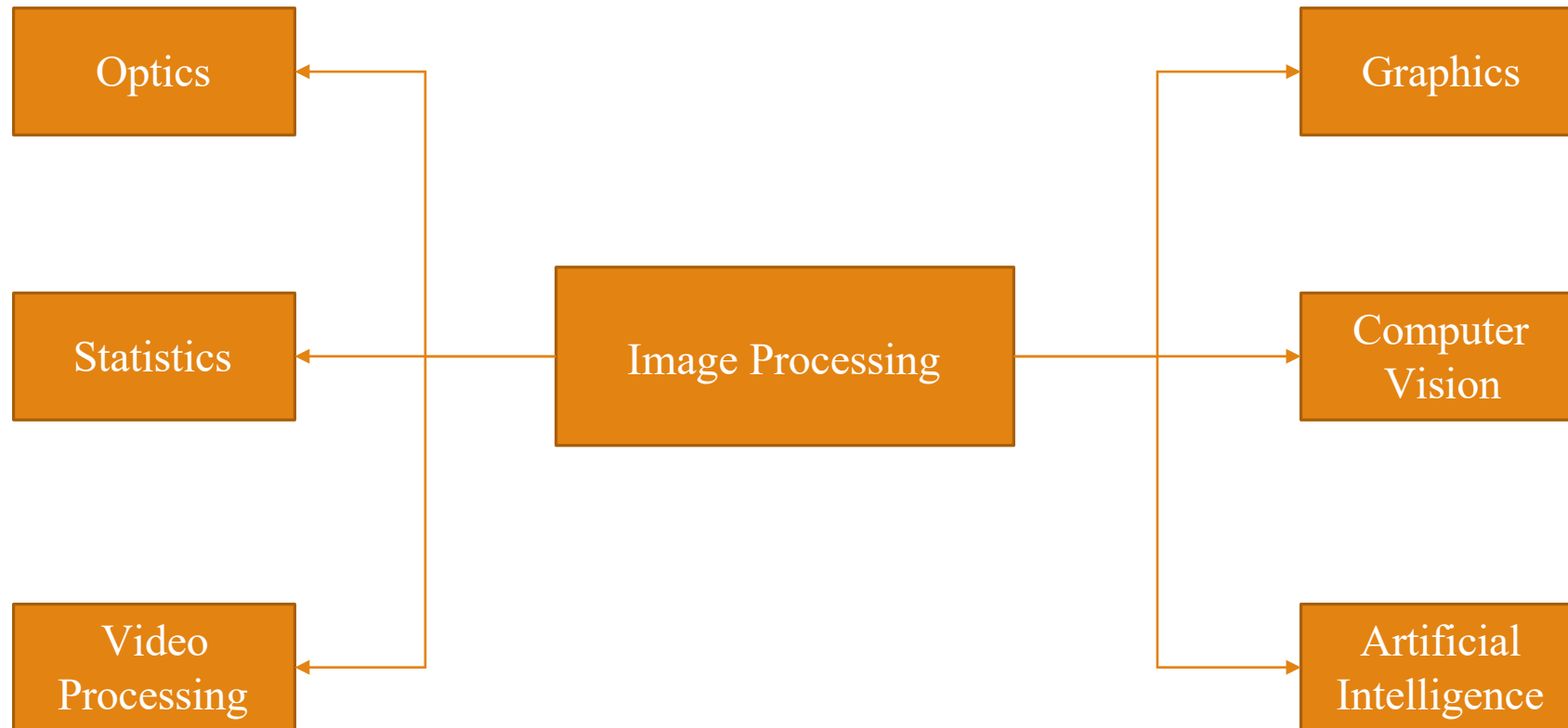
Challenges in Image Processing

The first major challenge in image processing is to acquire the image for further processing.

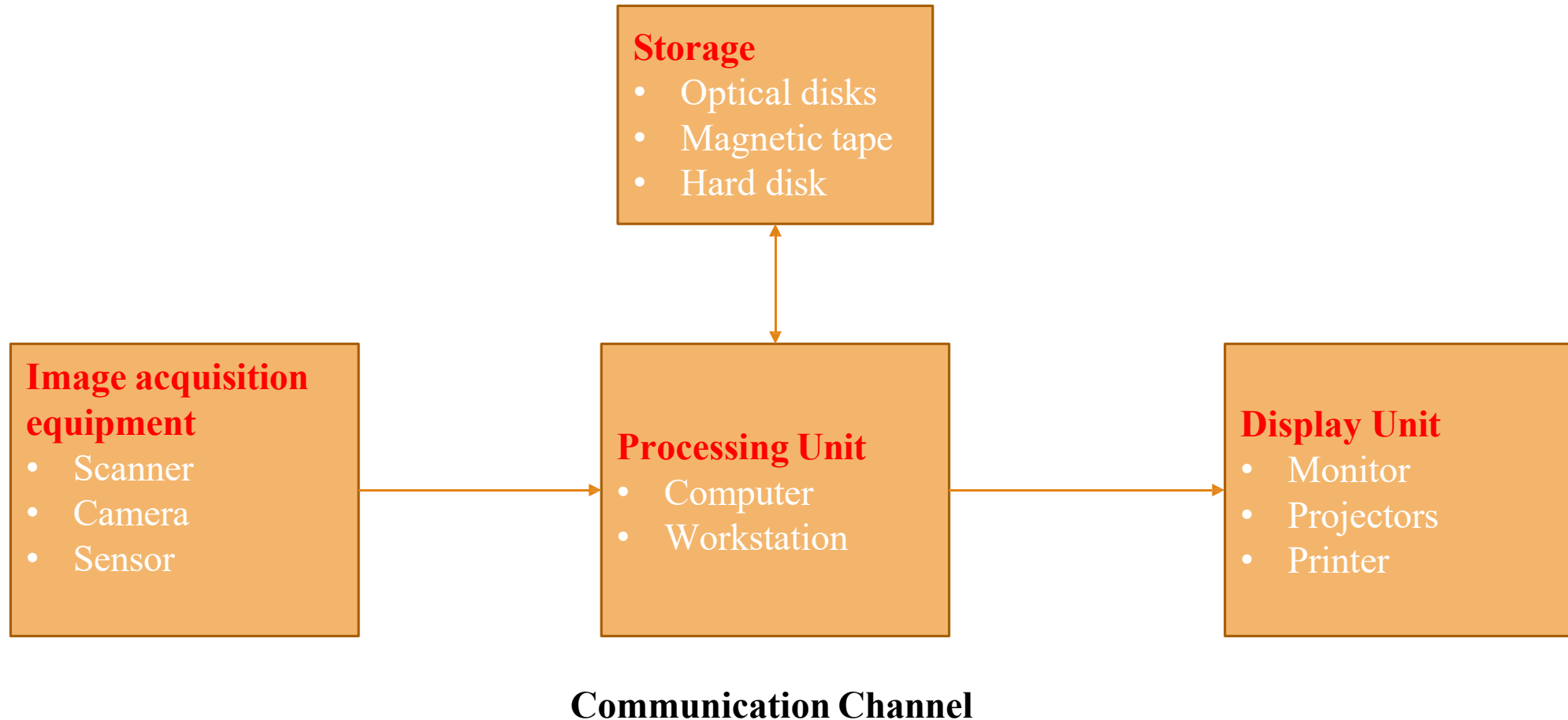
Three types of processing are there:

- 1. Optical image processing:** It is the study of radiation source, the object and the other optical processes involved.
- 2. Analog image processing:** It is an area that deals with the processing of analog electrical signals using analog circuits.
- 3. Digital image processing:** It is an area that uses digital circuits, systems, software algorithms to carry out the image processing tasks.

Image Processing and Related Fields



Elements of Digital Image Processing Systems



Fundamental Steps in Digital Image Processing

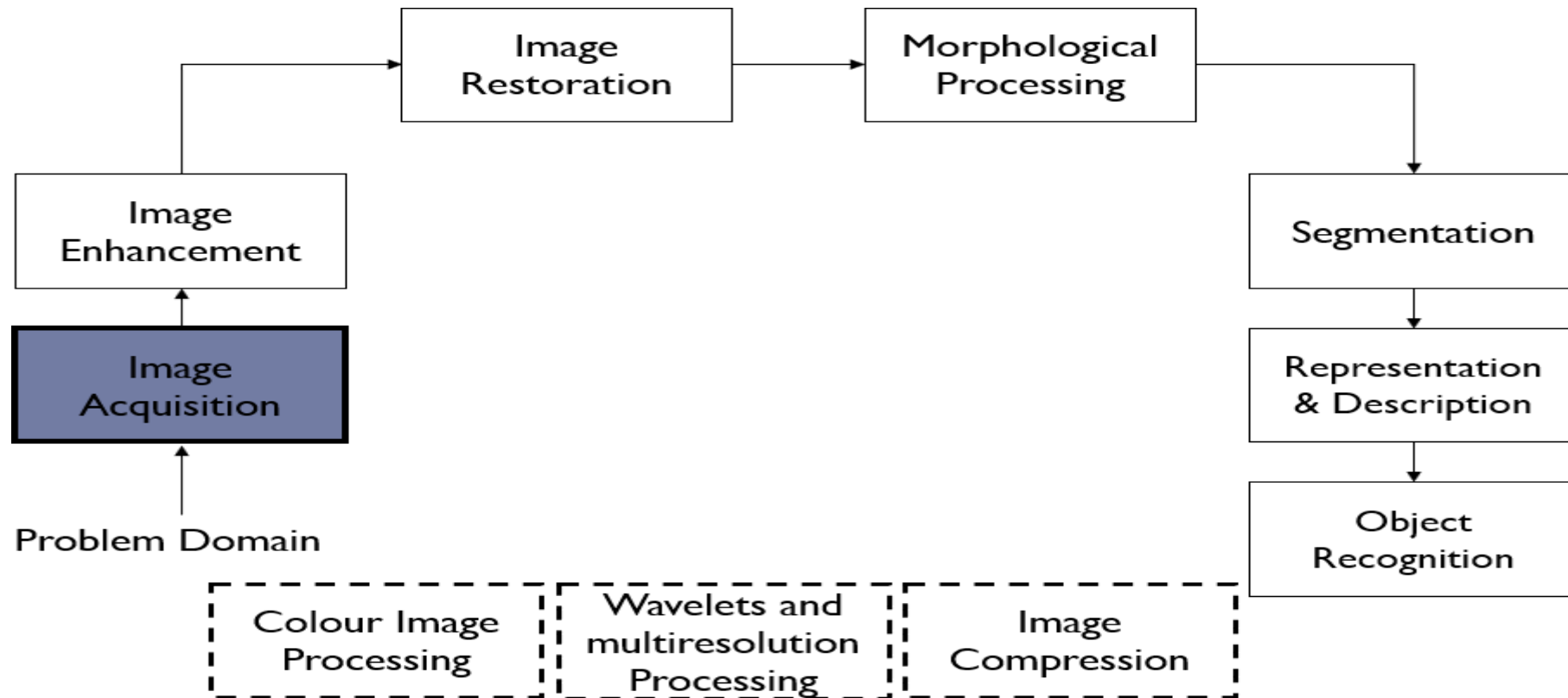


Figure 5: Steps of digital image processing

Image Acquisition

- ❖ Generally, the image acquisition stage involves preprocessing, such as scaling.

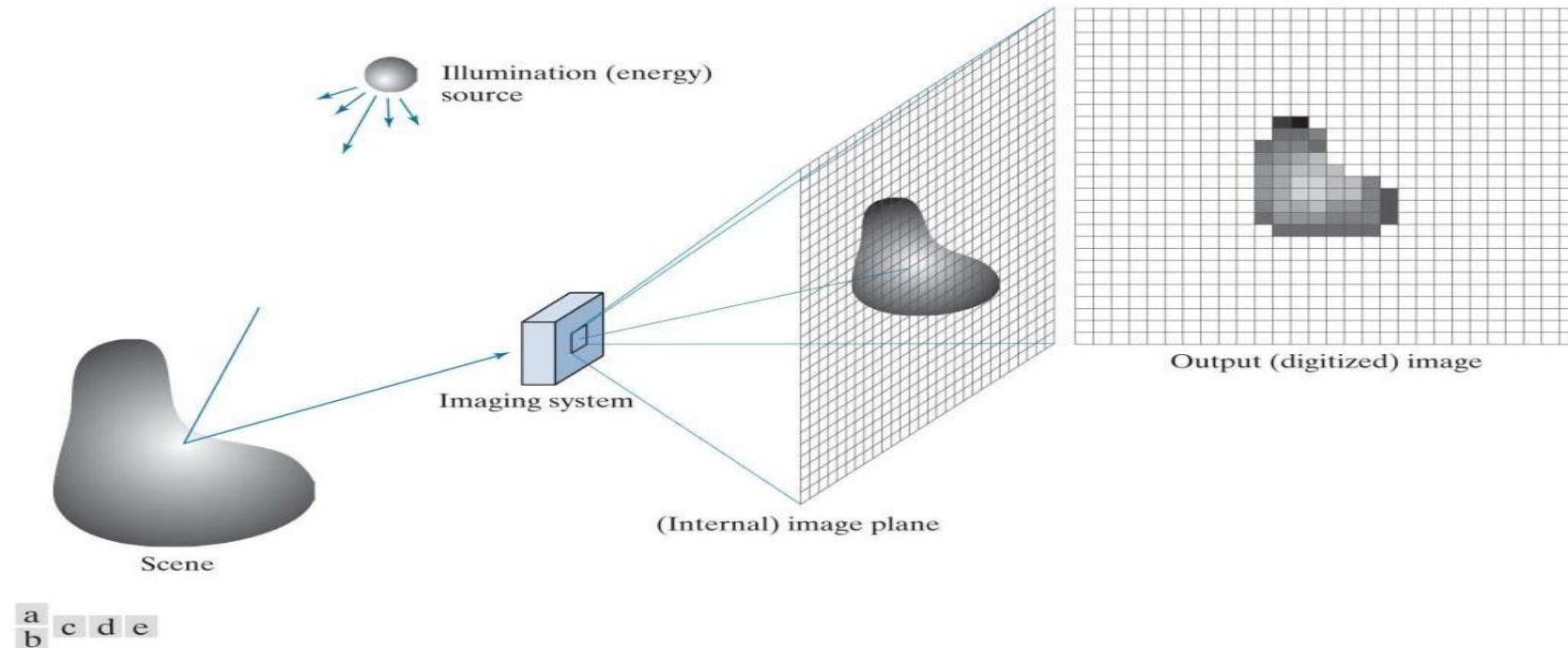


Figure 6: An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image Enhancement

- ❖ Image enhancement is the process of manipulating an image so that the result is more suitable than the original for a specific application.
- ❖ It involves brightness, contrast etc.

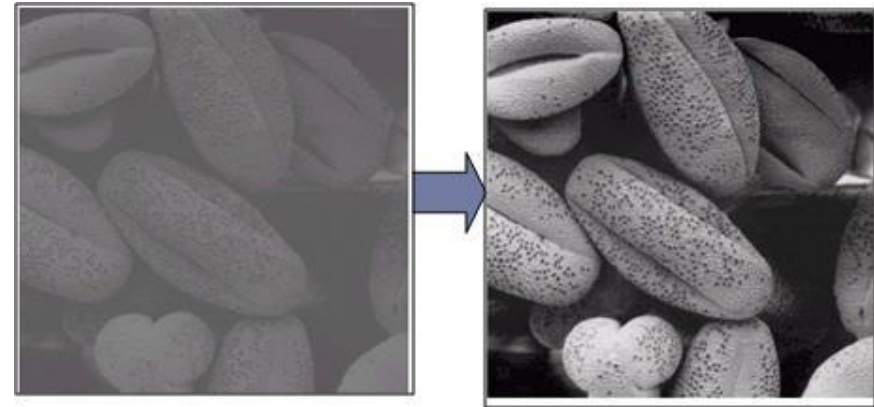


Figure 7: Image enhancement

Image Restoration

- ❖ Image restoration is an area that also deals with improving the appearance of an image.
- ❖ Unlike enhancement, which is subjective, image restoration is objective, in the sense that restoration techniques tend to be based on mathematical or probabilistic models of image degradation.
- ❖ It takes a corrupt/noisy image.
- ❖ Enhancement, on the other hand, is based on human subjective preferences regarding what constitutes a "good" enhancement result.

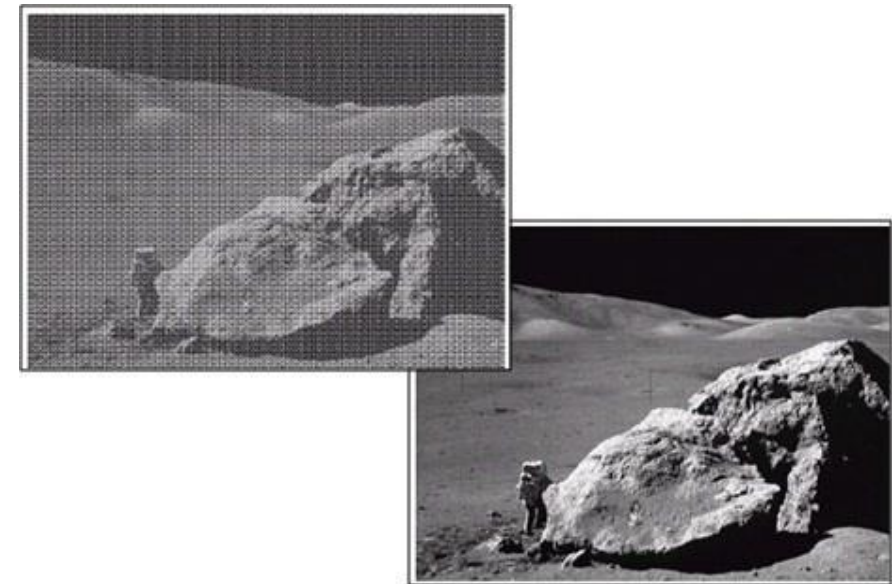


Figure 7: Image restoration

Color Image Processing

- ❖ Color image processing is an area that has been gaining in importance because of the significant increase in the use of digital images over the Internet.

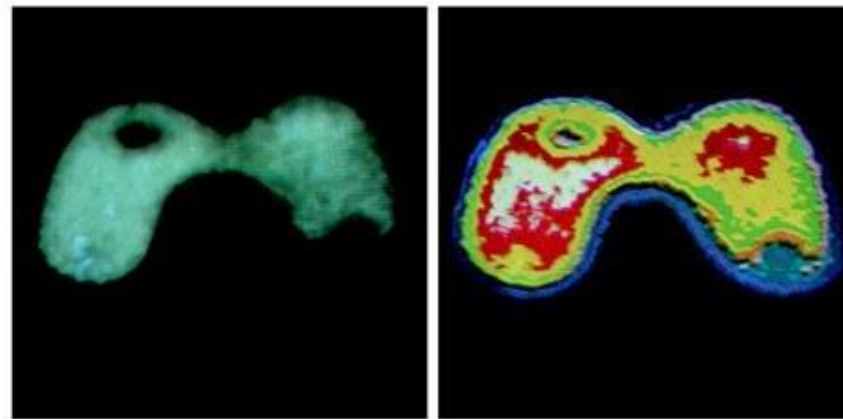


Figure 8: Color image processing

Morphological Processing

- ❖ Morphological processing deals with tools for extracting image components that are useful in the representation and description of shape.

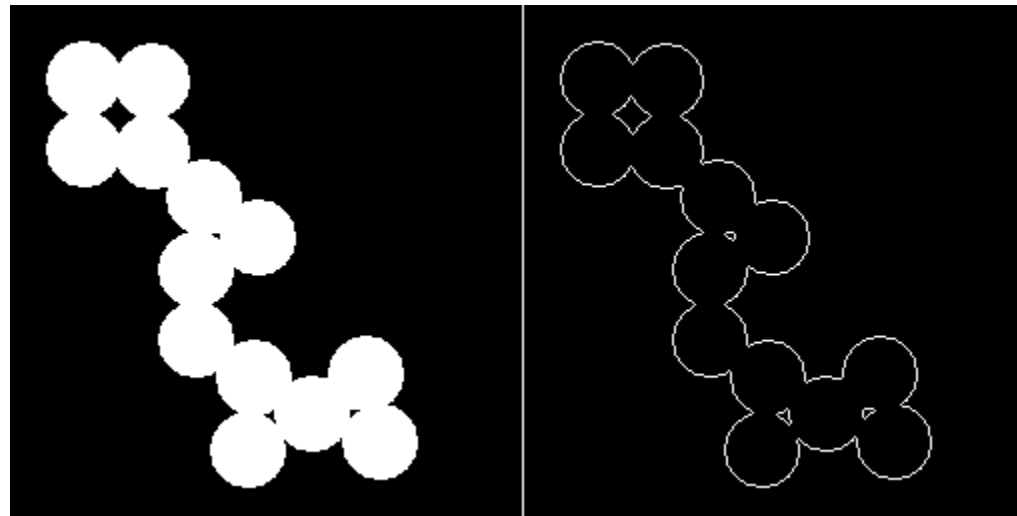


Figure 9: Morphological processing

Segmentation

- ❖ Segmentation procedures partition an image into its constituent parts or objects.

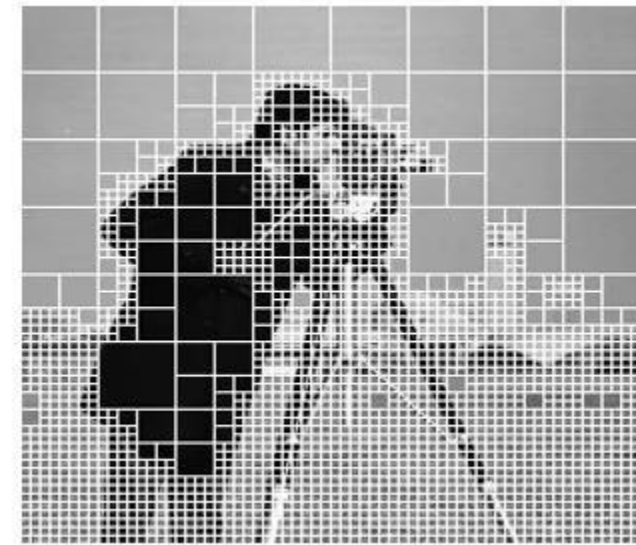


Figure 10: Image segmentation

Object Recognition

- ❖ Recognition is the process that assigns a label (e.g., "vehicle") to an object based on its descriptors.

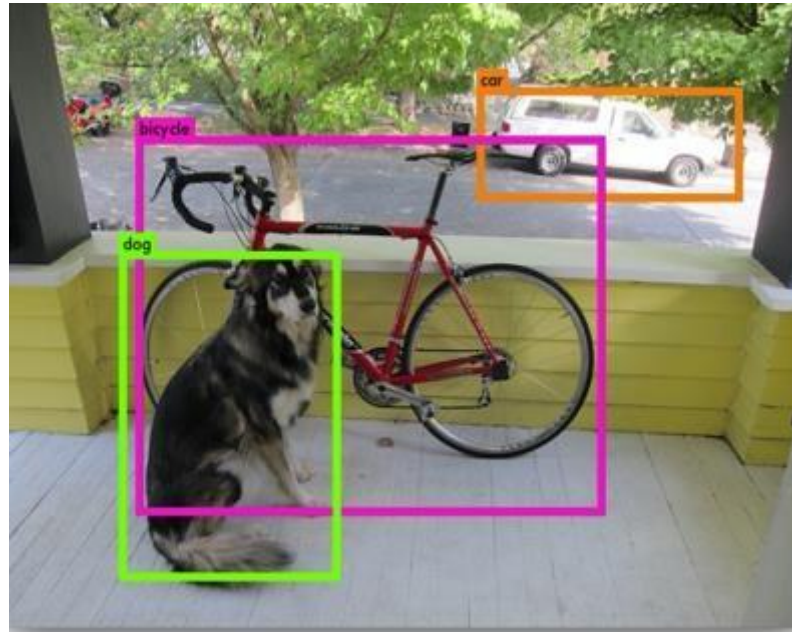


Figure 11: Object recognition

Image Compression

❖ Compression, as the name implies, deals with techniques for reducing the storage required to save an image, or the bandwidth required to transmit it.

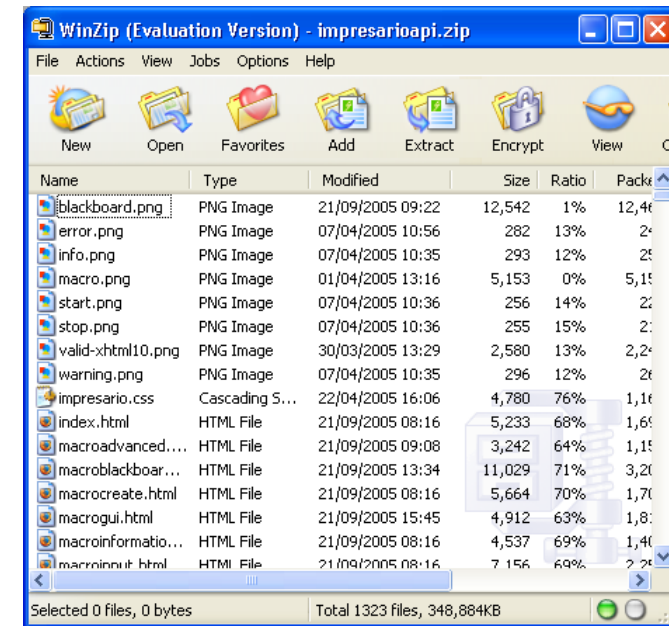


Figure 14: Image compression

Components of an Image Processing System

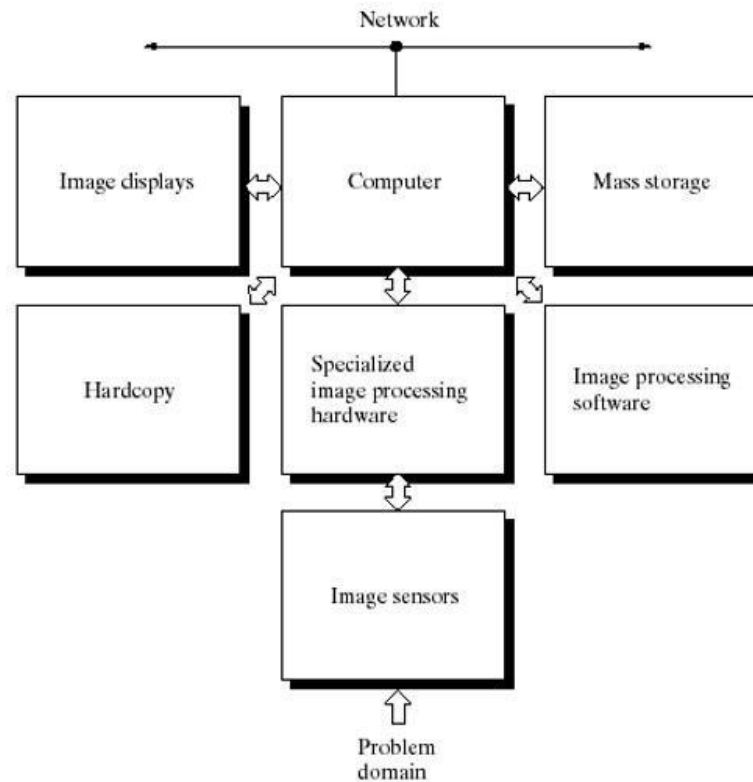
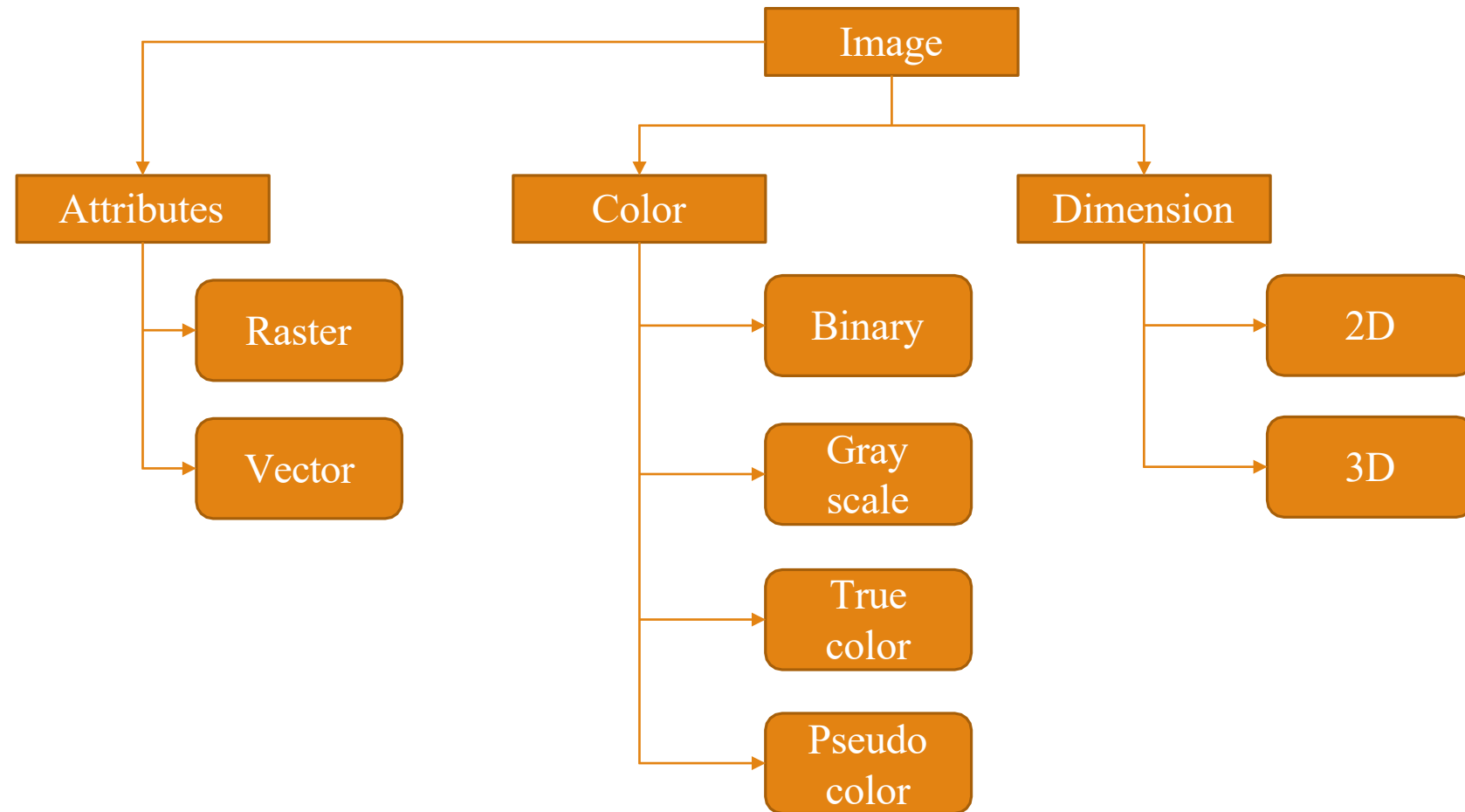


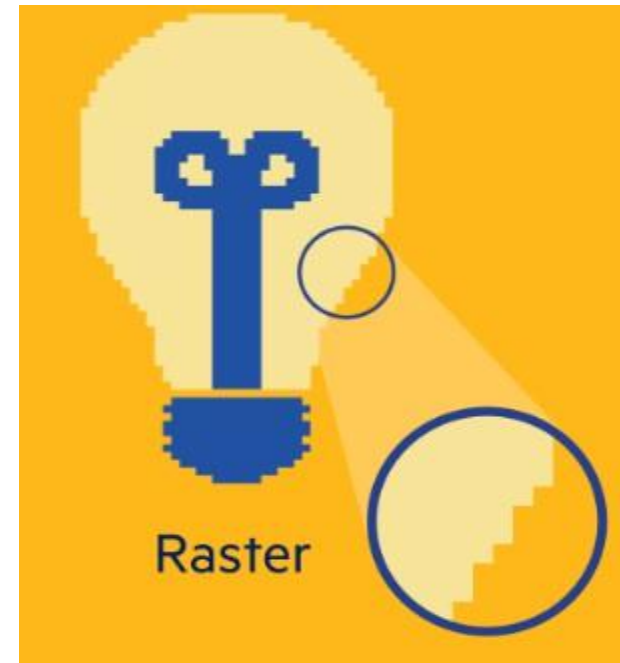
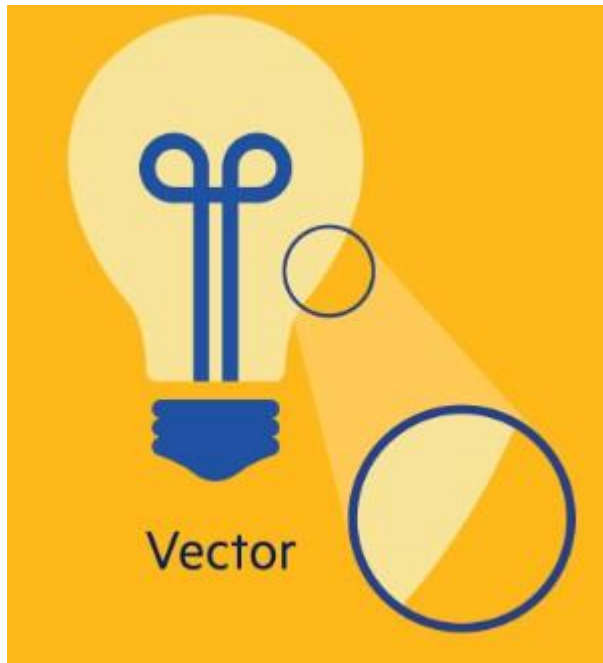
Figure 15: Components of image processing system

Image Types



Types of Images based on Attributes

- ❖ **Raster Images** are pixel based. The quality of the raster images depend on number of pixels.
- ❖ **Vector Graphics** use geometric attributes like lines and circles to describe an image.



Types of Images based on Color

- ❖ **Binary Images** are also called bi-level images. The pixel assume a value of 0 and 1.
- ❖ **Grey level Images** Eight bits ($2^8=256$) are enough to represent grey scale as human visual system can distinguish only 32 different levels. A gray scale image is a kind of black and white or gray monochrome are composed exclusively of shades of gray.
- ❖ **True Color Images** Also known as full color images represents the full range of available colors. Mostly they use 24 bits to represent all the colors. Hence, they can be considered as three band images.
- ❖ **Pseudo Color Images** These are false color images, where colors are added artificially based on the interpretation of data. They are popular in medical domain. Doppler color images are the example of pseudo color images.

Types of Images based on Dimensions

- ❖ **Generally** digital images are 2D rectangular array of pixels.
- ❖ If another dimensions are added like depth or any other characteristics, it is necessary to use higher order stack of images like 3D.

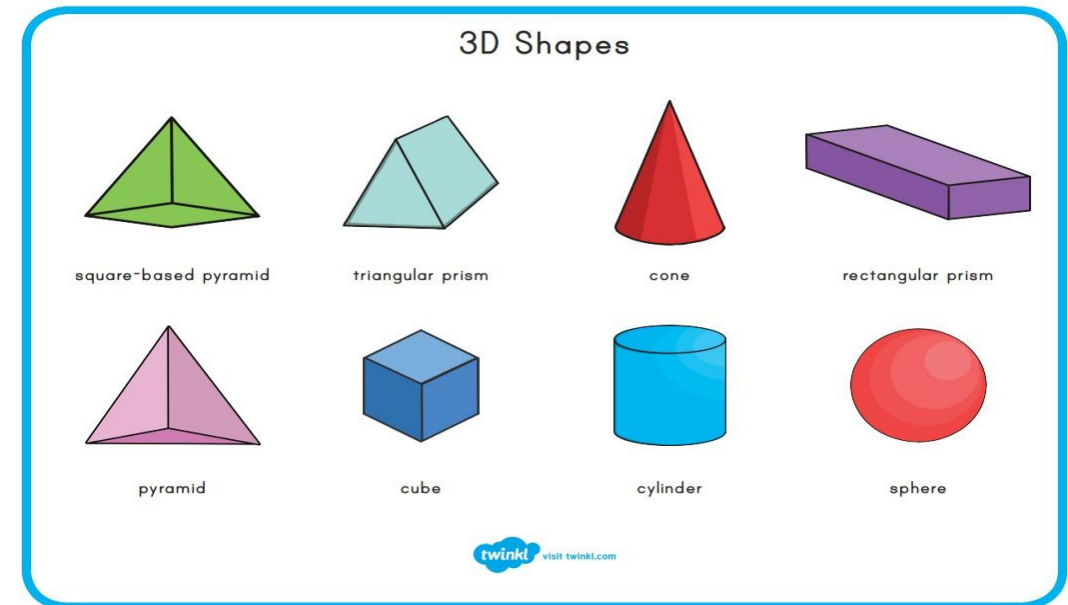
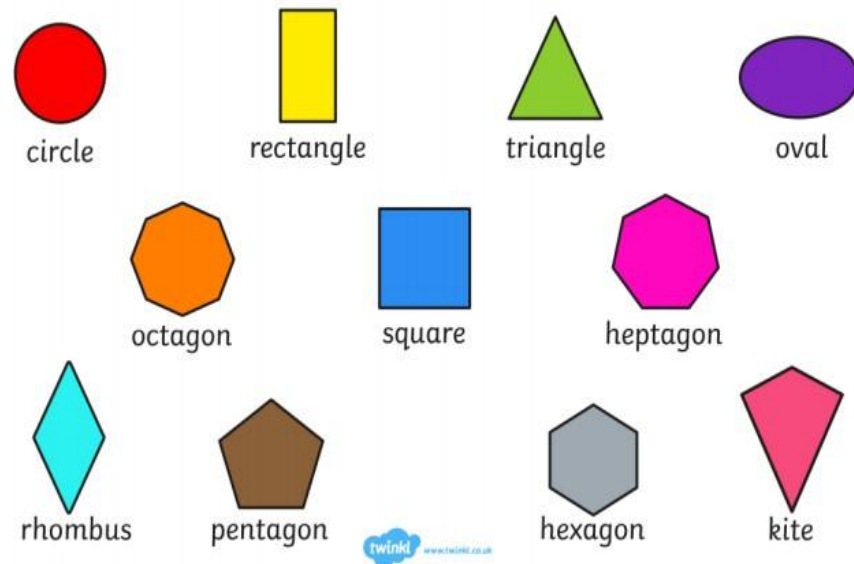


Image Formats

- ❖ **JPEG:** This is known for **Join Photographic Expert Group**. JPEGs are known for their "lossy" compression, meaning that the quality of the image decreases as the file size decreases. JPEG files are very common on the Internet and JPEG is a popular format for digital cameras - making it ideal for web use and non-professional prints.
- ❖ **GIF:** This is known as Graphics Interchange Format (Lossless compression). In their more basic form, GIFs are formed from up to 256 colors in the RGB color-space.
- ❖ **BMP:** This is known as Bit Map Picture. BMP or Bitmap Image File is a format developed by Microsoft for Windows. There is no compression or information loss with BMP files which allow images to have very high quality, but also very large file sizes.

Image Formats

- ❖ **TIFF:** TIFF or Tagged Image File Format are lossless images files meaning that they do not need to compress or lose any image quality or information (although there are options for compression), allowing for very high-quality images but also larger file sizes. **Compression:** Lossless - no compression. Very high-quality images. **Best For:** High quality prints, professional publications, archival copies.
- ❖ **PNG:** PNG or Portable Network Graphics files are a lossless image format originally designed to improve upon and replace the gif format. PNG files are able to handle up to 16 million colors, unlike the 256 colors supported by GIF. **Compression:** Lossless - compression without loss of quality **Best For:** Web Images.
- ❖ **EPS:** An EPS or Encapsulated PostScript file is a common vector file type. EPS files can be opened in many illustration applications such as Adobe Illustrator or CorelDRAW. **Compression:** None - uses vector information **Best For:** Vector artwork, illustrations

Thank You 😊
